QUASI-AFFINE MORPHISMS AND ALL IMMERSIONS ARE IND-QUASI-AFFINE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 19605-19609 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 19611-19618, the complete author proof (6 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Examples card of the ind-quasi-affine series; sequenced after both its inputs (dictionary L16109, composition L16116).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Definitions of ind-quasi-affine and quasi-affine morphisms; chapter lemma ind-quasi-affine-composition (card L16116, on the shelf).

Source reference. Original lemma 0F1V: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 19585–19587. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Any quasi-affine morphism is ind-quasi-affine. Any immersion is ind-quasi-affine.*

Proof. The first assertion is immediate from the definitions. In particular, affine morphisms, such as closed immersions, are ind-quasi-affine. Thus, by Lemma 0F1V, it remains to show that an open immersion is ind-quasi-affine. This, however, is immediate from the definitions. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.